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PREDICTIVE MODELING FOR BROWNFIELD MIGRATION FROM SAP ECC 6.0 TO S4HANA A MACHINE LEARNING APPROACH FOR EFFORT EVALUATION UNDER THE SAP RISE PROGRAM

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ABSTRACT

Migrating from SAP ECC 6.0 to SAP S/4HANA is one of the most important transformations in enterprise resource planning (ERP) modernization. This study examines the challenges and opportunities associated with brown field migration under the SAP RISE program, focusing on data size, customization, integration complexity, and estimated migration effort. Using a dataset of 25 migration events, descriptive and correlation analyses reveal strong relationships between database size, number of Z custom objects, and migration timelines. To improve predictability, Ada Boost Regression, Support Vector Regression (SVR), and Linear Regression were used as machine learning models to estimate migration effort. Linear regression and SVR demonstrated high accuracy with high R^2 values (>0.98) and low error margins, effectively modeling the migration complexity across various project sizes. Ada Boost, while powerful in handling nonlinearities, showed reduced generalizability, emphasizing the need for parameter tuning. The findings highlight that database size alone is not sufficient to assess the complexity of migration; customization and integration efforts significantly shape timelines and resource allocation. By combining statistical insights with predictive modeling, this research provides a structured framework for planning, optimizing, and executing SAP S/4HANA migrations, with reduced risks and improved performance. This research adds to the expanding collection knowledge on ERP modernization, providing IT managers, system architects, and business leaders with practical decision-support tools to navigate the complexities of digital transformation.

Keywords: SAP ECC 6.0, SAP S/4HANA Migration, Rise with SAP, Brownfield Approach, Machine Learning, Regression Models, Predictive Analytics, ERP Modernization.

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1. INTRODUCTION

SAP S/4HANA Cloud, with its advanced capabilities in real-time analytics, streamlined operations, and process automation, provides a robust solution to address these demands. However, the complexity of enterprise IT environments characterized by heterogeneous legacy systems, high data volumes, and mission-critical operations makes migration a challenging endeavor. [2] Now companies are facing major changes regarding transformation, and it is very important for them to improve the migration from their old ERP system to the new one. ERP migration is seen as one of the major challenges of ERP, as it is a serious and complex process. [3] SAP ERP is a useful tool for storing data for various types of businesses. SAP is an integrated tool (software) that connects all other business processes. Whenever there is any movement in the business process system, the SAP business use renters the data into the system, and the system immediately updates the information. And the information is available to all business users and management, who can view the user data anywhere in the world. [4] SAP S/4HANA will create the new SAP core for the coming decades. It aims to address the challenges of organizations by providing 1) improved reporting functionality (agility and speed), and 2) improved process efficiency (e.g. faster monthly closing process, improved user experience). [5] The transition from legacy SAP ECC systems to the next-generation SAP S/4HANA platform represents a watershed moment for global enterprise organizations.

This migration goes beyond the boundaries of a typical technology upgrade and constitutes a fundamental transformation of the entire enterprise resource planning (ERP) ecosystem. [6] The data migration component of S/4HANA implementations introduces multifaceted challenges that go beyond technical considerations. Historical data accumulated in legacy systems often contains inconsistencies, duplication, and structural inconsistencies that become very apparent during migration planning. [7] One of the primary benefits of migrating to SAP S/4HANA is the ability to access real-time insights. SAP HANA uses its in-memory computing capabilities, powered by the SAP HANA database, to process large volumes of data in real-time. This helps organizations gain immediate access to accurate, up-to-date information for better decision-making, gain improved visibility into business operations, and achieve improved operational efficiency. [8] The first thing we need to understand is the difference between SAP on premise and SAPS/4HANA on cloud. SAP on premise offers a lot of customization, while SAP S/4HANA on cloud is more standardized and has fewer customization capabilities. [9] The migration leverages the backup and recovery functionality of SAP HANA DB as the main technique for transferring and protecting data, it uses a multi-phase migration strategy, which has shown a 96% reduction in data migration time compared to conventional methods. [10] SAP S/4HANA is available in on premise, cloud and hybrid deployments to provide customers with real choices. Successful customers traditionally define their mining business or mining profession here. They can differentiate themselves from their competitors and realize a competitive advantage. [11] The process of transferring data from a legacy system - the original system - to a new system is called migration.

The new system incorporates the ERP business processes that are already in place functionality. [12] The convergence of several technological and methodological advancements has transformed SAP system modernization approaches.

The increasing adoption of DevOps methodologies has revolutionized implementation processes, with research showing that organizations that implement comprehensive DevOps practices in SAP environments achieve 67.4% faster release cycles and demonstrate 44.8% fewer production incidents after major changes. [13] SAP PO therefore provides a comprehensive solution for organizing business processes. Process Integration (SAP PI) integrates process integration, which is used to implement the interfaces mentioned in the cited text. [14] Organizations implementing RISE with SAP should conduct thorough analyses of how cloud infrastructure, business process intelligence, and integrated services directly contribute to strategic priorities such as market responsiveness, operational efficiency, and customer experience improvement. [15] RISE, powered by SAP, provides a comprehensive path to modernize legacy ERP systems by migrating to an AI-enabled cloud ERP environment built on SAP S/4HANA.

This approach helps organizations optimize their ERP landscapes and unlock the full potential of cloud-native technologies, ultimately driving better business outcomes. [16] Initially, the project will recruit five startups that offer products that complement SAP's existing product offering. In the best-case scenario, this collaboration will identify a group of customers whose needs can be met by integrating SAP's product with the participating company, opening the door to integration and commercialization. [17] In 2015, the company launched a product called SAP S/4HANA to replace previous generations of ERP software, as it is based on the SAP HANA in-memory database platform used to build the underlying databases used in each version of the software developed. SAP it increased the range of products it offers by adding supply chain management, human capital management, and cloud-based solutions customer relationship management. [18] The respondent was required to indicate the SAP modules and SAP support tools used in the implementation of this unit. The inputs were measured on a five-point Likert scale ranging from “Normal” to “In-depth”. The data collected in this section provided insights into the detailed design of the curriculum and the most popular SAP solutions.

2. MATERIALS AND METHODS

Materials:

ECC Data Size GB: Indicates the total size of the SAP ECC 6.0 database, measured in gigabytes, including all transactional, primary, and historical data. This value is very important for migration planning because it directly impacts system changeover time, infrastructure requirements, and data management strategies during the migration to SAP S/4HANA.

Number of Z Custom Objects: Indicates the total number of custom-developed objects within the SAP ECC system, usually prefixed with a “Z”. These include programs, tables, enhancements, and workflows. The number determines the level of customization complexity, code adaptation, and testing required during the migration process to SAP S/4HANA, ensuring operational balance and optimization.

Number of Interfaces: Indicates the total number of integration points, connections, or data exchange mechanisms between SAP ECC and other systems. These interfaces can use IDocs, APIs, middleware, or other technologies. The number helps in estimating the migration complexity as each interface needs to be tested, rebuilt or redesigned to ensure smooth communication after migration.

Estimated Migration Effort Days: Indicates the planned effort, measured in person-days, required for migration activities including system modification, testing, troubleshooting and optimization. This value is calculated considering database size, custom enhancements and interface complexity, providing a planning metric for resource allocation, budget and timeline estimation in SAP S/4HANA brownfield migration projects.

Instructions for Machine Learning

Linear Regression: A statistical method called linear regression models the relationship between a dependent variable and one or more independent variables by fitting a straight line to observed data. It works on the premise that the input and output variables have a linear relationship, which aims to predict outcomes. The method works by minimizing the difference between predicted results and actual observations using techniques such as least squares. It is widely used in economics, biology, social sciences, and business forecasting to analyze trends, make predictions, and determine the strength of predictors.

Support Vector Regression: A potent machine learning method is support vector regression, especially for regression tasks. Its strengths include identifying support vectors, determining the optimal hyper plane, and efficiently handling high-dimensional data, making it a very versatile and useful tool. Known as "support vectors," the algorithm in SVR chooses important data points to define a hyper plane. The main goal is to maximize the margin between these vectors and minimize boundary violations. In essence, SVR seeks to place the hyper plane as near to the desired data points as feasible without going overboard.A multidimensional surface known as a hyper plane is utilized in SVR to represent the relationship between data points in feature space and to generate predictions. It looks as a straight line when there are two characteristics, but in more complicated situations, it transforms into a higher-dimensional surface. The basic notion is always the same, regardless of complexity: SVR seeks to identify the best hyper plane for precise predictions.It is possible to describe the compression problem as a typical quadratic programming task and solve it using regular quadratic programming techniques. These techniques are computationally intensive, though, particularly when the Gram matrix is too big to store in memory. A more effective substitute that cuts down on computing time and avoids memory overflow is the decomposition method.

Ada Boost Regression: Ensemble modeling was transformed by Freud and Schapiro’s 1997 introduction of the Adaboost method. It has been widely applied to binary classification issues ever since. Adaboost transforms weak learners into strong and dependable models by combining many of them to increase prediction accuracy. AdaBoosting's basic concept is to train a model on a dataset initially, then iteratively create more models to account for the mistakes in the original ones. Until the prediction accuracy increases, this process keeps going. By combining several weak learners into a single, stronger model known as a "strong learner," AdaBoosting enhances performance. AdaBoost for predictive modelling is one various machine learning techniques are available problem statements. Adaptive boosting, or AdaBoost, is a group method that enhances model performance. It is referred to as "adaptive" because it modifies the weights, giving misclassified occurrences greater weight in order to increase accuracy.

3. RESULTS AND DISCUSSION

TABLE 1.SAP ECC 6.0 to SAP S/4HANA Migration (Brownfield – SAP RISE Program)

ECC Data Volume GB	Z Custom Objects Count	Interfaces Count	Estimated Migration Effort Days
672	3891	268	149
4677	2462	77	169
3979	4950	135	199
2717	3308	169	163
2683	2705	180	152
5172	2746	26	168
652	1425	196	108
4229	625	64	134
1328	3363	263	161
700	5336	242	166
3229	457	323	173
5857	5161	263	261
4454	4979	135	206
4602	1718	339	216
4347	3819	150	185
4748	1058	122	158
3152	4567	318	214
899	4227	137	129
5062	2178	37	159
2784	482	170	136
3077	5826	280	218
2319	2718	76	131
1217	5367	168	150
5571	4093	55	195
4722	4688	244	231

Table 1 provides a detailed overview of the key parameters that affect the migration of SAP ECC 6.0 systems to SAP S/4HANA under the SAP RISE program. Four key metrics are analyzed: ECC data size (GB), number of Z custom objects, number of interfaces, and estimated migration effort (days). Together, these parameters explain how database size, system customization, and integration complexity drive project planning and resource allocation. ECC data size ranges from small systems with a size of approximately 652 GB to large-scale landscapes exceeding 5,800 GB. Systems with larger data sizes, such as those greater than 5 TB, require more effort due to increased data cleansing, migration runtime, and infrastructure scalability. However, data size alone does not fully dictate project complexity. The number of Z custom objects, which represents the number of custom-built programs and enhancements, varies between 457 and 5,800 objects.

A higher number indicates significant customization, requiring more extensive code tuning, testing, and optimization to ensure compatibility with SAP S/4HANA's simplified data model. Similarly, an interface counts of 26 to 339 highlights the diversity of system connectivity. High interface counts indicate a broad integration landscape, requiring careful integration with external systems. The estimated migration effort demonstrates a clear correlation with these variables. Projects with large datasets, high customization, and complex integration—such as entries with more than 5,000 GB of data and more than 250 interfaces—require more than 200 person-days. In contrast, smaller, less complex systems are estimated to require around 100–130 days. This dataset underscores the need for tailored strategies for brownfield migrations. While database size affects runtime and infrastructure, customization and integration complexity are equally important in determining migration effort. These metrics help organizations effectively plan resources, prioritize remediation efforts, and optimize their transition to SAP S/4HANA under SAP RISE.

TABLE 2. Descriptive Statistics

	ECC Data Volume GB	Z Custom Objects Count	Interfaces Count	Estimated Migration Effort Days
Count	25	25	25	25
Mean	3313.96	3285.96	177.48	173.24
Std	1664.063863	1666.160568	93.437198	36.61202
Min	652	457	26	108
25%	2319	2178	122	150
50%	3229	3363	169	166
75%	4677	4688	263	199
Max	5857	5826	339	261

Table 2 provides descriptive statistics for the SAP ECC 6.0 to S/4HANA migration dataset, highlighting the prevalence and variability of key migration parameters. The dataset contains 25 observations, each representing a unique system migration scenario. The ECC data size averages approximately 3,314 GB, with a wide range from 652 GB to 5,857 GB, reflecting significant differences in system size and data complexity. Similarly, the number of Z custom objects shows considerable variability, with an average of 3,286 custom objects, but ranging from 457 to nearly 5,826, emphasizing varying levels of customization across systems.

The number of interfaces has a median of 177, with a minimum of 26 and a maximum of 339, suggesting that some landscapes are relatively simple while others involve extensive system integration. The estimated migration effort averaged 173 person-days, spanning between 108 and 261 days, reflecting how system size, customization, and integration complexity affect the resources required. The quarterly data also reveals clustering around medium to high values; for example, 75% of systems are larger than 2,319 GB in data size and 2,178 custom objects, indicating that most environments are large and complex.

Effect of Process Parameters

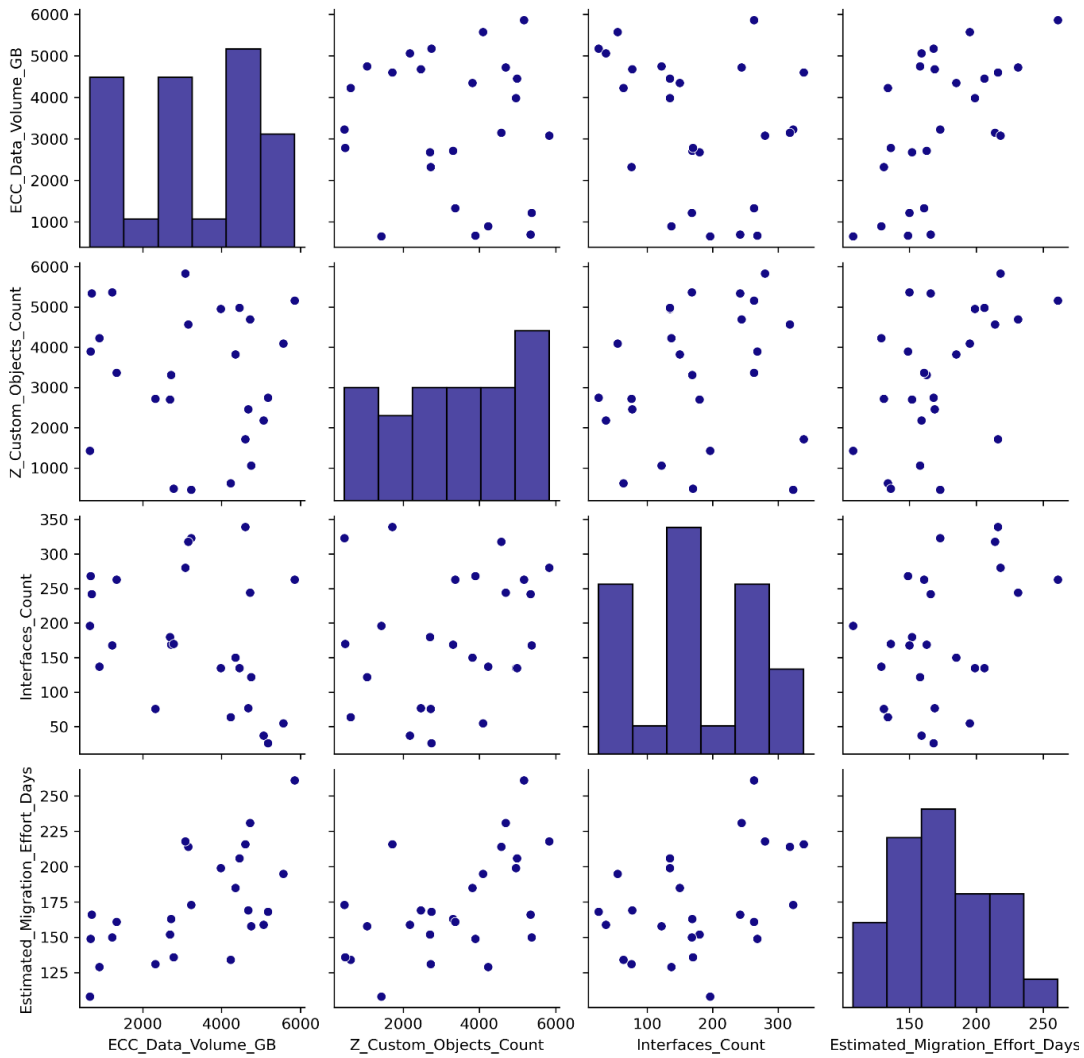


FIGURE 1. Scatter plot of the variousSAP ECC 6.0 to SAP S/4HANA Migration (Brownfield – SAP RISE Program)

Figure 1 presents a scatter plot matrix that visualizes the relationships between four important variables that affect the SAP ECC 6.0 to SAP S/4HANA brownfield migration under the SAP RISE program: ECC data volume (GB), number of Z custom objects, number of interfaces, and estimated migration effort (days). The diagonal histograms show the distribution of each parameter. ECC data volume, Z custom objects, and migration effort appear to be skewed to the right, indicating that while many settings fall within moderate ranges, some outliers with very high values significantly increase the migration complexity. The scatter plot reveals a generally positive relationship between all three input parameters of the migration effort. Larger database volumes demand more migration effort, although the relationship is not strictly linear due to variations in customization and integration complexity. Similarly, a larger number of custom objects and interfaces generally leads to longer migration timelines because additional tuning and testing are required. No strong correlation was found between the number of interfaces and ECC data volume, indicating that integration complexity is not necessarily linked to system size.

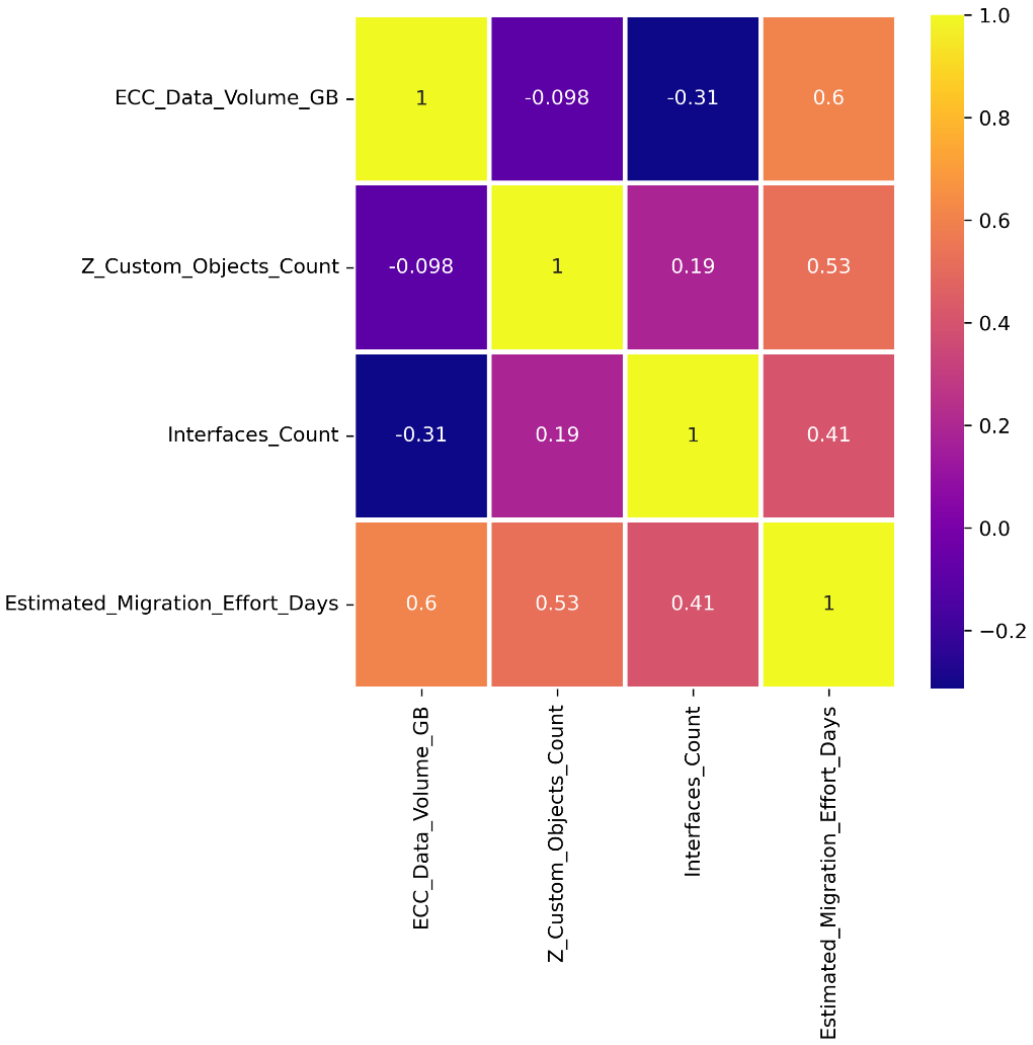


FIGURE 2. Correlation heat map on process parameters and effects

Figure 2 illustrates the correlations between the key variables that affect the SAP ECC 6.0 to SAP S/4HANA brownfield migration: ECC data volume (GB), number of Z custom objects, number of interfaces, and estimated migration effort (days). The color gradient reflects the strength and direction of the correlations, where values close to 1 indicate strong positive relationships, values close to -1 indicate strong negative relationships, and values around 0 indicate weak or no relationship. The most significant correlation is between ECC data volume and estimated migration effort (0.60), indicating that larger databases generally require more time and resources to migrate. Similarly, the number of Z custom objects shows a moderate positive correlation with migration effort (0.53), indicating that extensive system customization significantly impacts the project timeline. The number of interfaces also contributes to the migration complexity, with a moderate correlation of 0.41, indicating that system integration requests add to the overall effort. Interestingly, ECC data volume and number of interfaces exhibit a negative correlation (-0.31), indicating that system size does not necessarily determine the integration difficulty; smaller systems can still have a higher interface count.This heat map highlights that migration effort is a multifaceted outcome driven by database size, level of customization, and integration demands. This underscores the importance of comprehensive assessments rather than relying on a single metric when planning S/4HANA migrations.

Linear Regression

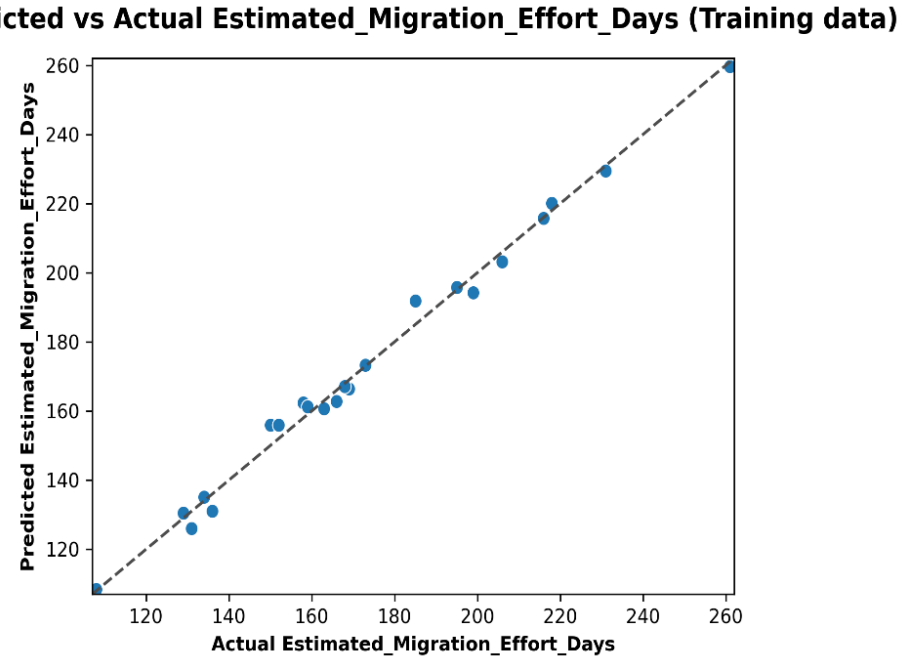


FIGURE 3. Linear Regression (Training data)

Figure 3 shows the performance of the trained linear regression model to predict the estimated migration effort (days) based on the ECC data volume (GB), the number of Z custom objects, and the number of interfaces. The scatter plot compares the predicted migration effort (y-axis) with the actual observed effort (x-axis) for the training dataset, with the dashed diagonal line representing a correct prediction. The data points are closely aligned with the diagonal, indicating that the model demonstrates strong predictive accuracy on the training data. This indicates a clear linear relationship between the three input parameters of migration effort, supporting the conclusion from previous analyses that larger data sizes, more customization, and complex integration collectively drive migration timelines. The minimal deviation from the diagonal line demonstrates that the model captures the underlying trends without significant underfitting or overfitting. The predictions are well distributed across the entire range, from low-effort projects (~120 days) to more complex migrations (>250 days), confirming its robustness across scenarios. This result highlights the effectiveness of linear regression for estimating SAP S/4HANA migration effort, making it a practical tool for project planning.

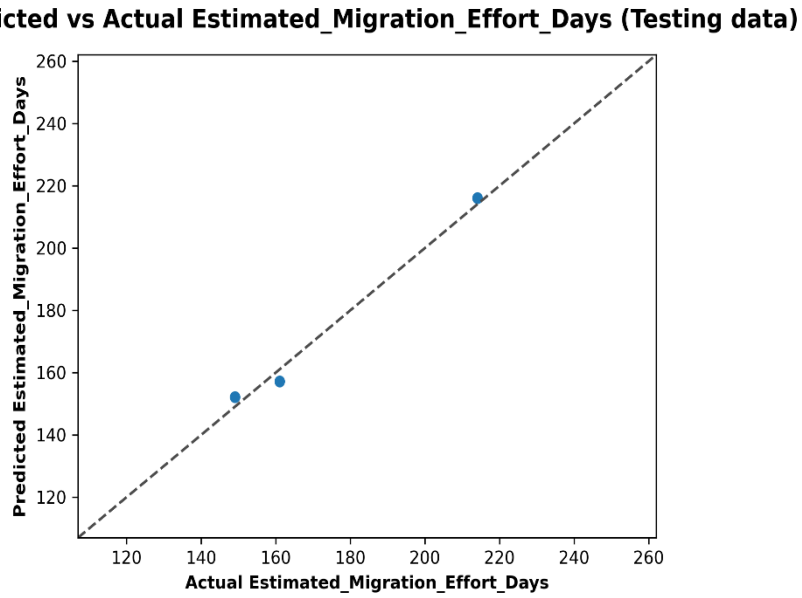


FIGURE 4. Linear Regression (Testing data)

Figure 4 illustrates the performance of the linear regression model on the test dataset, showing the predicted and actual estimated migration effort (days). The dashed diagonal line represents the correct prediction accuracy, while the scatter points represent the actual predictions for the missing data. The data points are very close to the diagonal, indicating that the model generalizes well to new cases beyond the training dataset. The predictions are accurate across different migration scenarios, with minimal error margins, demonstrating that the model effectively captures the relationship between migration effort and key drivers: ECC data volume, number of Z custom objects, and number of interfaces. Compared to the training data plot, there are fewer samples in the test data, however the high alignment with the reference line ensures consistency and reliability. The absence of large deviations or systematic bias indicates that the model avoids overfitting and provides robust performance when estimating project timelines. This performance makes the linear regression model a valuable decision-support tool for SAP S/4HANA brownfield migration planning. Organizations can confidently use it for preliminary effort assessment, resource allocation, and budget forecasting. However, with larger datasets and additional factors (e.g., infrastructure complexity, business downtime requirements), continuous refinement can further improve forecast accuracy and decision-making efficiency.

TABLE 3. Performance Metrics of Linear Regression (Training Data, Testing Data)

Data	Symbol	R2	EVS	MSE	RMSE	MAE	MaxError	MSLE	MedAE
Train	LR	0.99202	0.99202	10.79520	3.28560	2.68374	6.83566	0.00042	2.28572
Test	LR	0.98782	0.98813	9.71725	3.11725	3.03852	3.81229	0.00037	3.17882

Table 3 evaluates the performance of the linear regression model to predict the estimated migration effort (days). On the training data, the model achieves an R^2 of 0.992, showing an excellent fit with low error values (RMSE 3.29 days, MAE 2.68 days). The experimental results are similarly robust, with an R^2 of 0.988 and minimal errors (RMSE 3.12 days, MAE 3.04 days), confirming high generalization ability. The low MSLE and MaxError also indicate accuracy and consistency. These results demonstrate that linear regression effectively models migration effort, providing a reliable tool for planning SAP S/4HANA brownfield migration timelines and resource allocation.

Support Vector Regression

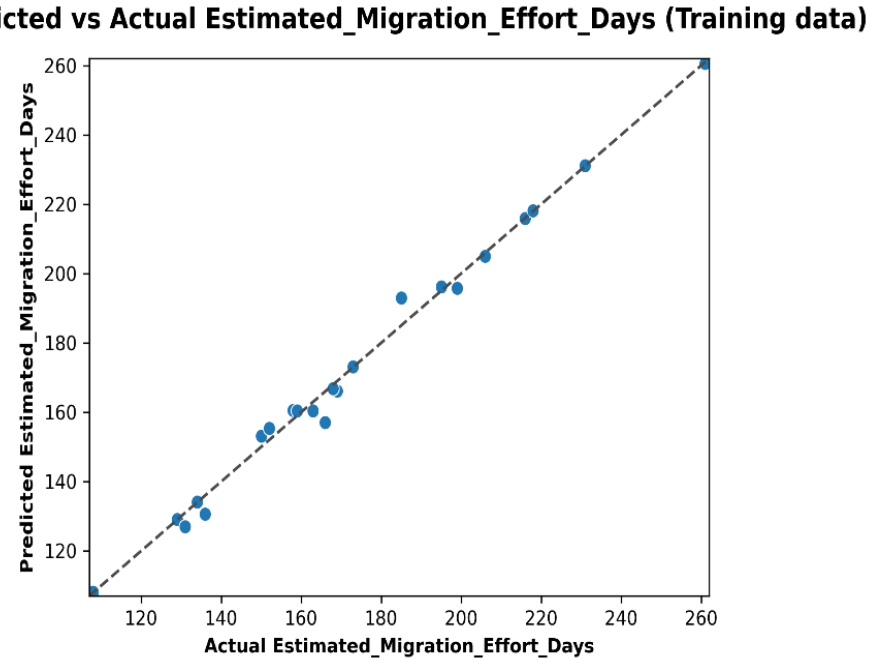


FIGURE 5. Support Vector Regression (Training data)

Figure 5 illustrates the performance of the Support Vector Regression (SVR) model for predicting estimated migration effort (days) on the training dataset. The scatter plot compares the actual and predicted values, with the dashed diagonal line indicating perfect prediction accuracy. Most of the data points are very close to the diagonal, indicating that the SVR model fits the training data exceptionally well and accurately captures the relationships between migration effort and key drivers: ECC data volume, number of Z custom objects, and number of interfaces. Compared to linear regression (Figure 3), SVR demonstrates similar or slightly tighter alignment, indicating that it effectively models the underlying nonlinear patterns in the dataset while maintaining low training error. A set of predictions on the diagonal for low- and high-effort migration projects shows that the SVR model handles a variety of project problems, from small systems requiring ~120 days to large-scale projects requiring over 250 days. The robust training performance indicates that SVR is capable of modeling the subtle interactions between parameters, making it suitable for complex SAP S/4HANA migration scenarios where customization and integration factors may not follow perfectly linear relationships.

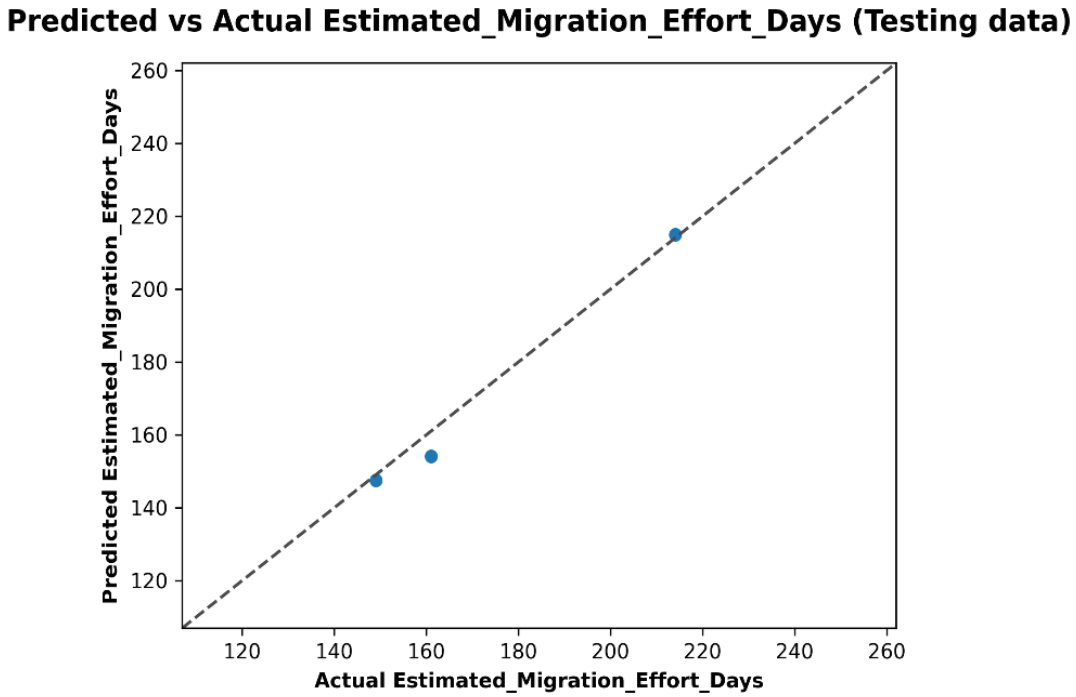


FIGURE 6. Support Vector Regression (Testing data)

Figure 6 demonstrates the performance of the Support Vector Regression (SVR) model on the test dataset, comparing the predicted and actual estimated migration effort (days). The points closely follow the diagonal line, indicating strong prediction accuracy and good generalization to missing data. The predictions for both the low-effort and high-effort projects agree well with the observed values with minimal deviation. This indicates that the SVR effectively captures the complex relationships between data size, customization, and interface complexity without overestimating them. The consistency of the model across the training and test datasets confirms its robustness, making SVR a reliable tool for estimating SAP S/4HANA migration effort in various project environments.

TABLE 4. Performance Metrics of Support Vector Regression (Training Data, Testing Data)

Data	Symbol	R2	EVS	MSE	RMSE	MAE	MaxError	MSLE	MedAE
Train	SVR	0.99171	0.99183	11.22393	3.35021	2.26916	8.88093	0.00043	1.36558
Test	SVR	0.97910	0.98671	16.67231	4.08317	3.05468	6.87466	0.00066	1.40479

Table 4 evaluates the performance of the Support Vector Regression (SVR) model to predict the estimated migration effort (days). On the training data, the model achieves $R^2 = 0.992$, showing an excellent fit with low errors (RMSE ≈ 3.35 days, MAE ≈ 2.27 days). The test performance is robust ($R^2 = 0.979$), confirming good generalization despite slightly higher errors (RMSE ≈ 4.08 days). The low MSLE and MedAE values indicate accuracy and consistency across all predictions. These results demonstrate the ability of SVR to model complex relationships between migration parameters, making it a robust approach for accurate SAP S/4HANA migration effort estimation across diverse organizational landscapes.

AdaBoost Regression

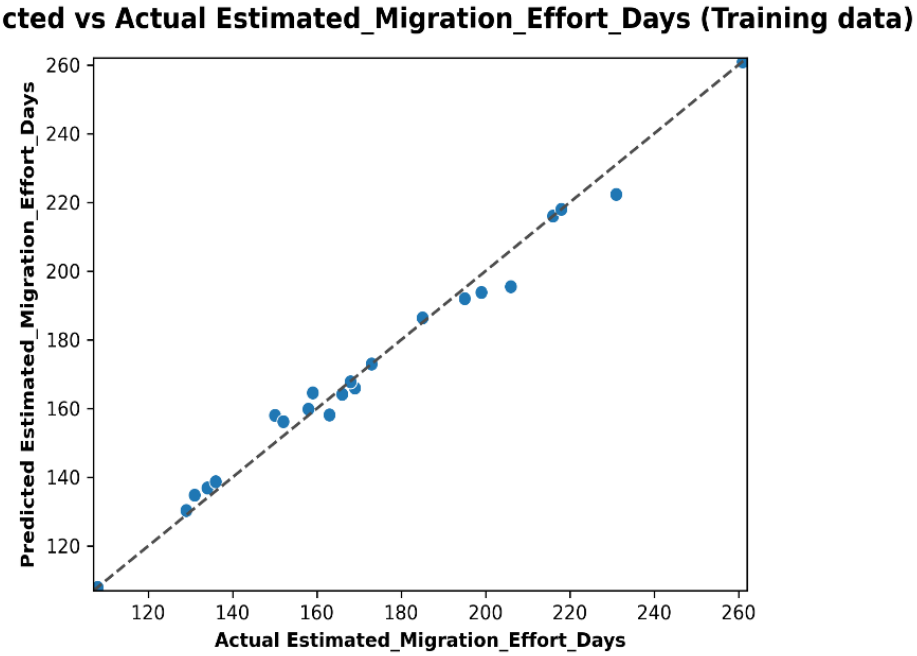


FIGURE 7. Ada Boost Regression (Training data)

Figure 7 illustrates the performance of the AdaBoost regression model in predicting the estimated migration effort (days) on the training dataset. The scatter plot compares the predicted values with the actual observations, with the dashed diagonal line indicating the correct prediction accuracy. Most of the data points are clustered closely along this line, reflecting high prediction accuracy and a strong fit to the training data. AdaBoost, an ensemble learning method, combines multiple weak learners (typically decision trees) to create a robust prediction model that can capture complex, nonlinear relationships. The figure shows that AdaBoost effectively models the relationships between the ECC data volume, the number of Z custom objects, and the number of interfaces with minimal visible deviation, even for projects with high migration efforts of more than 220 days. Compared to linear models, AdaBoost excels at handling parameter interactions and variance, which is valuable in SAP S/4HANA migration projects where complexity does not follow strict linear trends. However, the tight alignment in the training data also implies the need to evaluate performance on the test data to ensure generalization and prevent overfitting.

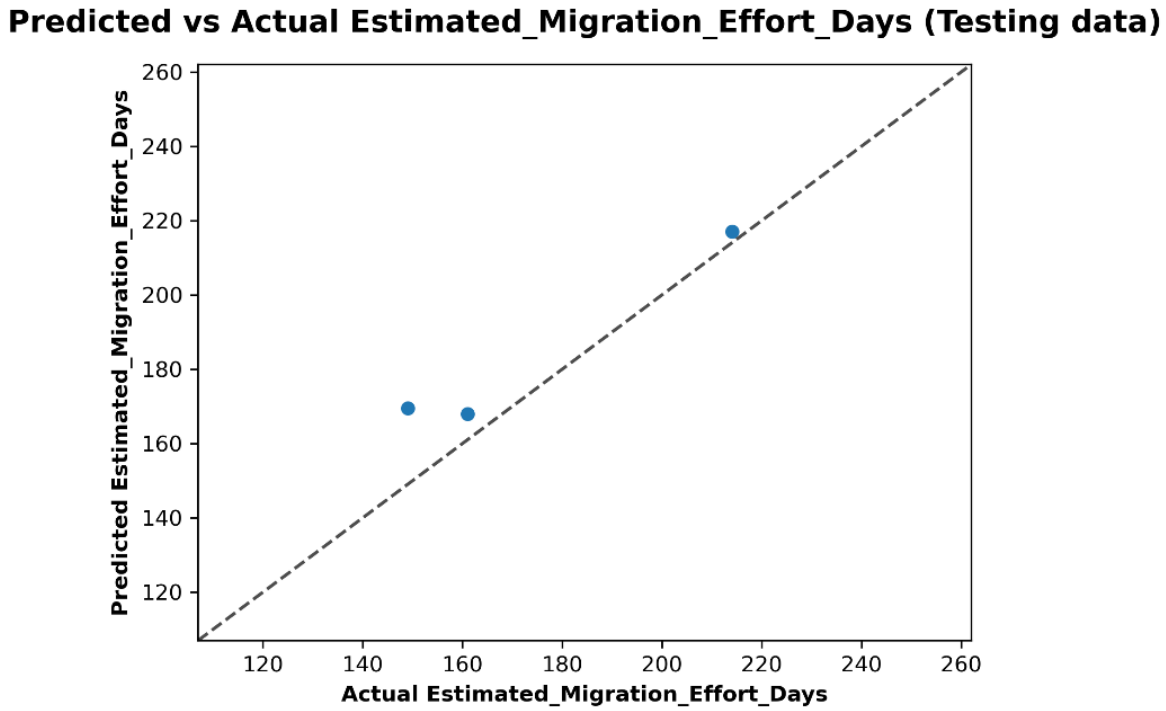


FIGURE 8. Ada Boost Regression (Testing data)

Figure 8 evaluates the AdaBoost regression model on the test dataset, comparing predicted and actual estimated migration effort (days). The data points are reasonably aligned with the diagonal line, indicating solid predictive performance, although the predictions show slightly higher variance compared to the training data. Despite the small sample size, AdaBoost demonstrates strong generalization, effectively estimating migration timelines for unseen projects. The model's ability to capture nonlinear parameter interactions across data size, customization, and integration complexity makes it valuable for planning SAP S/4HANA migrations. Small deviations highlight the need for tuning and more training data, but overall, AdaBoost remains a robust, adaptive predictive modeling approach.

TABLE 5. Performance Metrics of AdaBoost Regression (Training Data, Testing Data)

Data	Symbol	R2	EVS	MSE	RMSE	MAE	MaxError	MSLE	MedAE
Train	ABR	0.98635	0.98638	18.47643	4.29842	3.14190	10.50000	0.00059	2.90000
Test	ABR	0.80012	0.92972	159.41667	12.62603	10.16667	20.50000	0.00613	7.00000

Table 5 summarizes the performance of AdaBoost regression (ABR) for predicting estimated migration effort (days). On the training data, the model achieves strong accuracy ($R^2 \approx 0.99$, RMSE ≈ 4.30 days, MAE ≈ 3.14 days), indicating an excellent fit. However, the test performance shows a significant drop ($R^2 \approx 0.80$, RMSE ≈ 12.63 days, MAE ≈ 10.17 days), indicating reduced generalization and possible overfitting. The high MaxError and MedAE values further confirm the variability in predictions for missing data. Although AdaBoost effectively models nonlinear relationships, these results highlight the need for parameter tuning, additional data, or regularization to improve performance for practical migration planning scenarios.

4. CONCLUSION

A major turning point in improving enterprise resource planning systems was the transition from SAP ECC 6.0 to SAP S/4HANA, moving beyond a typical technology upgrade to a comprehensive transformation of business operations. The study demonstrated that migration complexity is affected by three critical parameters: database size, number of Z custom objects, and integration interfaces. While large data volumes directly extend runtime and infrastructure demands, customization intensity and integration landscape introduce additional complexity, often requiring significant code adaptation, testing, and optimization. As a result, organizations must consider these dimensions holistically rather than relying solely on data volume as a proxy for migration effort. The use of machine learning techniques provided valuable predictive insights for migration planning. Linear regression performed exceptionally well, achieving high predictive accuracy on both training and test data, making it a reliable baseline model for effort assessment. Support Vector Regression (SVR) further improved predictive power by effectively capturing nonlinear interactions, demonstrating robustness across different migration scenarios.

While Ada Boost Regression showed promising results during training, its experimental performance highlights limitations in generalization, underscoring the importance of parameter optimization and the availability of large datasets for improved reliability. Together, these models underscore the potential of predictive analytics to reduce uncertainty in project planning, improve resource allocation, and enable proactive risk management. From a practical perspective, organizations preparing for SAP S/4HANA migration should adopt a structured approach that integrates descriptive statistical analysis with predictive modeling. This ensures that effort estimates are grounded in empirical evidence, while being tailored to the unique challenges of each organizational landscape. Furthermore, migration planning should extend beyond technical considerations to include governance, data management, and change management, as these elements play a critical role in ensuring project success. The findings of this study highlight that successful SAP S/4HANA migrations require a balance of technical expertise, analytical foresight, and strategic decision-making. By leveraging machine learning models alongside traditional migration methods, organizations can improve prediction accuracy, reduce downtime, and optimize costs. This integrated approach not only supports smooth transitions, but also empowers organizations to unlock the full potential of SAP S/4HANA’s real-time analytics, automation, and cloud-enabled capabilities. Ultimately, adopting data-driven migration strategies will help organizations stay competitive in an increasingly dynamic digital landscape.

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